

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson		Centre Number	Candidate Number
Edexcel Award			
Wednesday 26 June 2019			
Morning (Time: 3 hours)		Paper Reference 9811/01	
Advanced Extension Award			
Mathematics			
You must have: Mathematical Formulae and Statistical Tables			Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible. e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. (a) By writing $u = \log_4 r$, where $r > 0$, show that

$$\log_4 r = \frac{1}{2} \log_2 r \quad (2)$$

- (b) Solve the equation

$$\log_4 (5x^2 - 11) = \log_2 (3x - 5) \quad (5)$$

(a) $u = \log_4 r$

$$4^u = r$$

TAKE LOG BASE 2 OF BOTH SIDES

$$\log_2 4^u = \log_2 r$$

$$u \log_2 4 = \log_2 r$$

$$2u = \log_2 r$$

$$u = \frac{1}{2} \log_2 r$$

$$\therefore \log_4 r = \frac{1}{2} \log_2 r$$

(b) $\log_4 (5x^2 - 11) = \log_2 (3x - 5)$

$$4^{\log_4 (5x^2 - 11)} = 4^{\log_2 (3x - 5)}$$

$$5x^2 - 11 = (2^2)^{\log_2 (3x - 5)}$$

$$5x^2 - 11 = 2^{2 \log_2 (3x - 5)}$$

$$5x^2 - 11 = 2^{\log_2 (3x - 5)^2}$$

$$5x^2 - 11 = (3x - 5)^2$$



Question 1 continued

$$5x^2 - 11 = 9x^2 - 30x + 25$$

$$4x^2 - 30x + 36 = 0$$

$$2x^2 - 15x + 18 = 0$$

$$(2x - 3)(x - 6) = 0$$

$$\therefore x = \frac{3}{2}, 6$$

$$\text{BUT } 3x - 5 < 0 \text{ WHEN } x = \frac{3}{2}$$

$$\therefore \log_2(3x - 5) \text{ IS UNDEFINED AT } x = \frac{3}{2}$$

$$\therefore x = 6 \text{ ONLY}$$

(Total for Question 1 is 7 marks)



2. The discrete random variable X follows the binomial distribution

$$X \sim B(n, p)$$

where $0 < p < 1$. The mode of X is m .

(a) Write down, in terms of m , n and p , an expression for $P(X = m)$

(1)

(b) Determine, in terms of n and p , an interval of width 1, in which m lies.

(5)

(c) Find a value of n where $n > 100$, and a value of p where $p < 0.2$, for which X has **two** modes. For your chosen values of n and p , state these two modes.

(2)

(a) $P(X = m) = \binom{n}{m} p^m (1-p)^{n-m}$

(b)

NOTE: $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

WANT $\underline{P(X = m-1) \leq P(X = m)}$ OR $\underline{P(X = m+1) \leq P(X = m)}$

$$P(X = m-1) = \binom{n}{m-1} p^{m-1} (1-p)^{n-(m-1)}$$

$$= \frac{n!}{(m-1)!(n-(m-1))!} p^{m-1} (1-p)^{n-m+1}$$

$$= \frac{n!}{(m-1)!(n-m+1)!} p^{m-1} (1-p)^{n-m+1}$$

$$P(X = m) = \binom{n}{m} p^m (1-p)^{n-m}$$

$$= \frac{n!}{m!(n-m)!} p^m (1-p)^{n-m}$$

$$P(X = m+1) = \binom{n}{m+1} p^{m+1} (1-p)^{n-(m+1)}$$



Question 2 continued

$$= \frac{n!}{(m+1)! \{n-(m+1)\}!} p^{m+1} (1-p)^{n-(m+1)}$$

$$= \frac{n!}{(m+1)! (n-m-1)!} p^{m+1} (1-p)^{n-m-1}$$

CONDITION 1:

$$\frac{n!}{(m-1)! (n-m+1)!} p^{m-1} (1-p)^{n-m+1} \leq \frac{n!}{m! (n-m)!} p^m (1-p)^{n-m}$$

$$\frac{p^{m-1} (1-p)^{n-m+1}}{(m-1)! (n-m+1)!} \leq \frac{p^m (1-p)^{n-m}}{m! (n-m)!}$$

$$\frac{p^{m-1} (1-p)^{n-m+1}}{(m-1)! (n-m+1)(n-m)!} \leq \frac{p^m (1-p)^{n-m}}{m(m-1)! (n-m)!}$$

$$\frac{p^{m-1} (1-p)^{n-m+1}}{n-m+1} \leq \frac{p^m (1-p)^{n-m}}{m}$$

$$\frac{p^m (1-p)^{n-m} (1-p)}{p} \leq \frac{p^m (1-p)^{n-m}}{m}$$

$$\frac{1-p}{p} \leq \frac{1}{m}$$

$$\frac{1-p}{n-m+1} \leq \frac{p}{m} \quad \textcircled{1}$$

CONDITION 2:

$$\frac{n!}{(m+1)! (n-m-1)!} p^{m+1} (1-p)^{n-m-1} \leq \frac{n!}{m! (n-m)!} p^m (1-p)^{n-m}$$

(Total for Question 2 is 8 marks)



P 6 0 7 0 2 A 0 5 2 8

$$\frac{p^{m+1} (1-p)^{n-m-1}}{(m+1)! (n-m-1)!} \leq \frac{p^m (1-p)^{n-m}}{m! (n-m)!}$$

$$\frac{p^{m+1} (1-p)^{n-m-1}}{(m+1)m! (n-m-1)!} \leq \frac{p^m (1-p)^{n-m}}{m! (n-m)(n-m-1)!}$$

$$\frac{p^{m+1} (1-p)^{n-m-1}}{m+1} \leq \frac{p^m (1-p)^{n-m}}{n-m}$$

$$\frac{\cancel{p^m} p \cancel{(1-p)^{n-m}}}{(1-p)} \leq \frac{\cancel{p^m} \cancel{(1-p)^{n-m}}}{n-m}$$

$$\frac{\frac{p}{1-p}}{m+1} \leq \frac{1}{n-m}$$

$$\frac{p}{m+1} \leq \frac{1-p}{n-m} \quad (2)$$

$$\text{From } (1): \frac{1-p}{n-m+1} \leq \frac{p}{m}$$

$$m(1-p) \leq p(n-m+1)$$

$$m - \cancel{mp} \leq pn - \cancel{mp} + 1$$

$$m \leq pn + p$$

$$\therefore m \leq p(n+1) \quad (3)$$

$$\text{From } (2): \frac{p}{m+1} \leq \frac{1-p}{n-m}$$

$$p(n-m) \leq (m+1)(1-p)$$

$$pn - \cancel{pm} \leq m+1 - \cancel{pm} - p$$

$$pn \leq m+1-p$$

$$pn + p \leq m+1$$

$$pn + p - 1 \leq m$$

$$p(n+1) - 1 \leq m \quad (4)$$

COMBINE (3) AND (4) INTO A SINGLE INEQUALITY:

$$\therefore p(n+1) - 1 \leq m \leq p(n+1)$$

(c) NOTE: CAN CHOOSE ANY VALUE OF n AND p SUCH THAT $n > 100$ AND $p < 0.2$ AND REQUIRE THE END POINTS OF THE INTERVALS OF LENGTH 1 TO BE INTEGERS
 $\therefore m$ MUST BE AN INTEGER

$$\text{TRY } p = 0.1 : 0.1(n+1) - 1 \leq m \leq 0.1(n+1)$$

$$\therefore 0.1(n+1) - 1 \text{ AND } 0.1(n+1) \text{ ARE INTEGERS}$$

$$\text{COULD HAVE } n = 109$$

$$\text{i.e. } 0.1(109+1) - 1 \leq m \leq 0.1(109+1)$$

$$9 \leq m \leq 10$$

$\therefore$ MODES COULD BE 9 AND 10

3. Given that $\phi = \frac{1}{2}(\sqrt{5} + 1)$,

(a) show that

(i) $\phi^2 = \phi + 1$

(ii) $\frac{1}{\phi} = \phi - 1$

(4)

(b) The equations of two curves are

$$y = \frac{1}{x} \quad x > 0$$

and $y = \ln x - x + k \quad x > 0$

where k is a positive constant.

The curves touch at the point P .

Find in terms of ϕ

(i) the coordinates of P ,

(ii) the value of k .

(6)

(+S1)

(a) $\phi = \frac{1}{2}(\sqrt{5} + 1)$

(i) LHS : $\phi^2 = \left\{ \frac{1}{2}(\sqrt{5} + 1) \right\}^2$

$$= \frac{1}{4}(\sqrt{5} + 1)(\sqrt{5} + 1)$$

$$= \frac{1}{4}(5 + 2\sqrt{5} + 1)$$

$$= \frac{1}{4}(6 + 2\sqrt{5})$$

$$= \frac{1}{2}(\sqrt{5} + 3)$$

RHS : $\phi + 1 = \frac{1}{2}(\sqrt{5} + 1) + 1$

$$= \frac{1}{2}(\sqrt{5} + 1) + \frac{2}{2}$$

$$= \frac{1}{2}(\sqrt{5} + 1 + 2)$$



Question 3 continued

$$= \frac{1}{2} (\sqrt{5} + 3)$$

AS LHS = RHS $\therefore \phi^2 = \phi + 1$

(ii)

$$\text{LHS : } \frac{1}{\phi} = \frac{1}{\frac{1}{2}(\sqrt{5}+1)}$$

$$= \frac{2}{\sqrt{5}+1} \left(\frac{\sqrt{5}-1}{\sqrt{5}-1} \right)$$

$$= \frac{2\sqrt{5} - 2}{5 - 1}$$

$$= \frac{2\sqrt{5} - 2}{4}$$

$$= \frac{1}{2} (\sqrt{5} - 1)$$

$$\text{RHS : } \phi - 1 = \frac{1}{2} (\sqrt{5} + 1) - 1$$

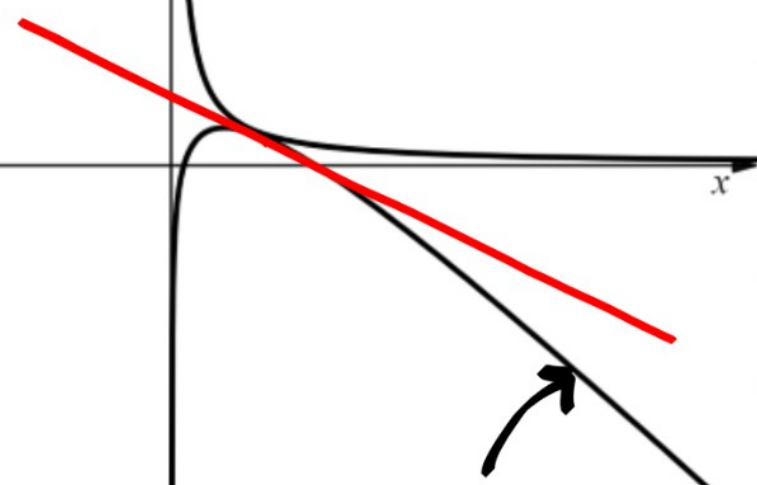
$$= \frac{1}{2} \sqrt{5} + \frac{1}{2} - 1$$

$$= \frac{1}{2} \sqrt{5} - \frac{1}{2}$$

$$= \frac{1}{2} (\sqrt{5} - 1)$$

AS LHS = RHS $\therefore \frac{1}{\phi} = \phi - 1$

$y = \frac{1}{x}$



$y = 4x - x + k$

NOTE : THE CURVES TOUCH WHEN THEIR DERIVATIVES ARE EQUAL; THEY HAVE A COMMON TANGENT



Question 3 continued

$$y = \frac{1}{x} \quad \therefore \frac{dy}{dx} = -\frac{1}{x^2}$$

$$y = \ln x - x + k \quad \therefore \frac{dy}{dx} = \frac{1}{x} - 1$$

$$\therefore -\frac{1}{x^2} = \frac{1}{x} - 1$$

$$-1 = x - x^2$$

$$x^2 - x - 1 = 0$$

$$\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} - \frac{4}{4} = 0$$

$$\left(x - \frac{1}{2}\right)^2 = \frac{5}{4}$$

$$x = \frac{1}{2} \pm \frac{1}{2}\sqrt{5}$$

$$\text{As } x > 0 \quad \therefore x = \frac{1}{2} + \frac{1}{2}\sqrt{5}$$

$$\therefore x = \frac{1}{2}(\sqrt{5} + 1) = \phi$$

$$\therefore x_p = \phi \Rightarrow y_p = \frac{1}{\phi} = \phi - 1$$

$$\therefore p(\phi, \phi - 1)$$

$$x = \phi, y = \phi - 1: \phi - 1 = \ln \phi - \phi + k$$

$$\therefore k = 2\phi - \ln \phi - 1$$



4. (a) Prove the identity

$$(\sin x + \cos y) \cos(x - y) \equiv (1 + \sin(x - y))(\cos x + \sin y) \quad (5)$$

(b) Hence, or otherwise, show that

$$\frac{\sin 5\theta + \cos 3\theta}{\cos 5\theta + \sin 3\theta} \equiv \frac{1 + \tan \theta}{1 - \tan \theta} \quad (6)$$

(c) Given that $k > 1$, show that the equation $\frac{\sin 5\theta + \cos 3\theta}{\cos 5\theta + \sin 3\theta} = k$ has a unique solution in the interval $0 < \theta < \frac{\pi}{4}$ (4)

(+S2)

(a) LHS : $(\sin x + \cos y) \cos(x - y)$

$$\begin{aligned} &\equiv (\sin x + \cos y)(\cos x \cos y + \sin x \sin y) \\ &\equiv \sin x \cos x \cos y + \sin^2 x \sin y + \cos x \cos^2 y + \sin x \cos y \sin y \\ &\equiv \sin x \cos x \cos y + (1 - \cos^2 x) \sin y + \cos x (1 - \sin^2 y) + \sin x \cos y \sin y \\ &\equiv \sin x \cos x \cos y + \sin y - \sin y \cos^2 x + \cos x - \cos x \sin^2 y + \sin x \cos y \sin y \\ &\equiv \cos x + \sin y + \cos x (\sin x \cos y - \cos x \sin y) + \sin y (\sin x \cos y - \cos x \sin y) \\ &\equiv \cos x + \sin y + \cos x \{\sin(x - y)\} + \sin y \{\sin(x - y)\} \\ &\equiv (\cos x + \sin y) \{1 + \sin(x - y)\} \\ &\equiv \text{RHS}, \text{ HENCE SHOWN} \end{aligned}$$



Question 4 continued

(b) LHS : $\frac{\sin 5\theta + \cos 3\theta}{\cos 5\theta + \sin 3\theta}$

LET $x = 5\theta$ AND $y = 3\theta$

$$\frac{\sin x + \cos y}{\cos x + \sin y}$$

FROM (a) : $(\sin x + \cos y) \cos(x-y) \equiv \{1 + \sin(x-y)\}(\cos x + \sin y)$

$$\frac{\sin x + \cos y}{\cos x + \sin y} \equiv \frac{1 + \sin(x-y)}{\cos(x-y)}$$

$$\frac{1 + \sin(x-y)}{\cos(x-y)}$$

$$\frac{1 + \sin(5\theta - 3\theta)}{\cos(5\theta - 3\theta)}$$

$$\frac{1 + \sin 2\theta}{\cos 2\theta}$$

$$\frac{\cos^2 \theta + \sin^2 \theta + 2\sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta}$$

$$\frac{\cos^2 \theta + 2\sin \theta \cos \theta + \sin^2 \theta}{\cos^2 \theta - \sin^2 \theta}$$

$$\frac{(\cos \theta + \sin \theta)^2}{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}$$

$$\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$$



Question 4 continued

$$\frac{\cos \theta}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$$

$$\frac{\cos \theta}{\cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$\frac{1 + \tan \theta}{1 - \tan \theta}, \text{ HENCE SHOWN}$$

(c) $\frac{\sin 5\theta + \cos 3\theta}{\cos 5\theta + \sin 3\theta} = k$

$$\frac{1 + \tan \theta}{1 - \tan \theta} = k$$

$$1 + \tan \theta = k(1 - \tan \theta)$$

$$1 + \tan \theta = k - k \tan \theta$$

$$\tan \theta + k \tan \theta = k - 1$$

$$\tan \theta (1 + k) = k - 1$$

$$\tan \theta = \frac{k-1}{k+1}$$

NOTE :

$$\begin{array}{r} 1 \\ k+1 \overline{) k-1} \\ \underline{k+1} \\ -2 \end{array}$$

$$\tan \theta = 1 - \frac{2}{k+1}$$

$$\text{AS } k \rightarrow \infty : \tan \theta \rightarrow 1 - 0$$



Question 4 continued

$$\tan \theta \rightarrow 1$$

$$\therefore \tan \theta < 1$$

$$\text{As } k > 1, k+1 > 2 \therefore 1 - \frac{2}{k+1} > 0 \therefore 0 < \tan \theta$$

$$\therefore 0 < \tan \theta < 1$$

$$\therefore \tan^{-1} 0 < \theta < \tan^{-1} 1$$

$$\therefore 0 < \theta < \frac{\pi}{4}$$

(Total for Question 4 is 17 marks)



5. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively, relative to an origin O , and are such that OAB is a triangle with $OA = a$ and $OB = b$.

The point C , with position vector $\mathbf{c}$, lies on the line through O that bisects the angle AOB .

- (a) Prove that the vector $b\mathbf{a} - a\mathbf{b}$ is perpendicular to $\mathbf{c}$. (4)

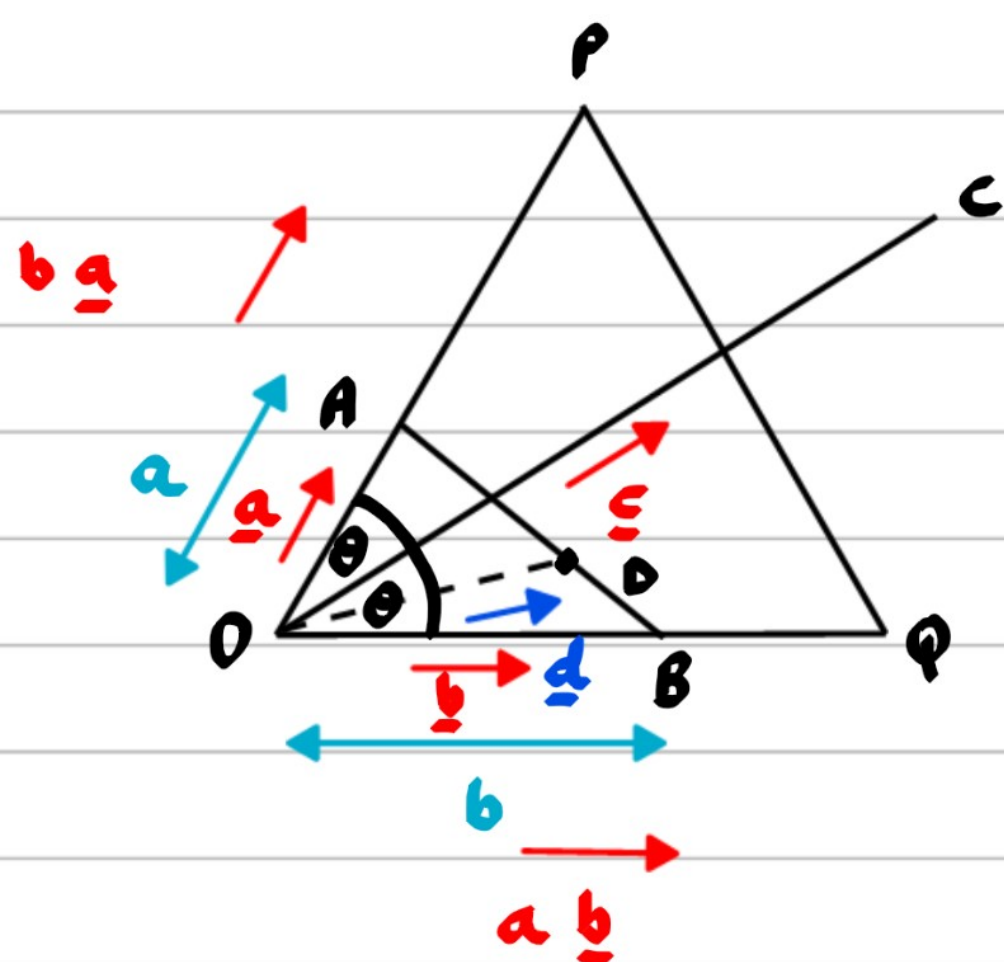
The point D , with position vector $\mathbf{d}$, lies on the line AB between A and B .

- (b) Explain why $\mathbf{d}$ can be expressed in the form $\mathbf{d} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}$ for some scalar λ with $0 < \lambda < 1$ (2)

- (c) Given that D is also on the line OC , find an expression for λ in terms of a and b only and hence show that

$$DA : DB = OA : OB \quad (8)$$

(+S2)



LET P AND Q BE THE POINTS SUCH THAT $\vec{OP} = b\mathbf{a}$ AND $\vec{OQ} = a\mathbf{b}$ RESPECTIVELY

i.e. $|\vec{OP}| = b|\mathbf{a}| = ba$ AND $|\vec{OQ}| = a|\mathbf{b}| = ab$

$\therefore |\vec{OP}| = |\vec{OQ}|$

$\therefore OPQ$ IS ISOSCELES

$\therefore$ ANGLE BISECTOR FROM O IS PERPENDICULAR TO PQ

$\vec{QP} = \vec{OP} - \vec{OQ} = b\mathbf{a} - a\mathbf{b}$

Question 5 continued

AS C IS ON THE PERPENDICULAR BISECTOR

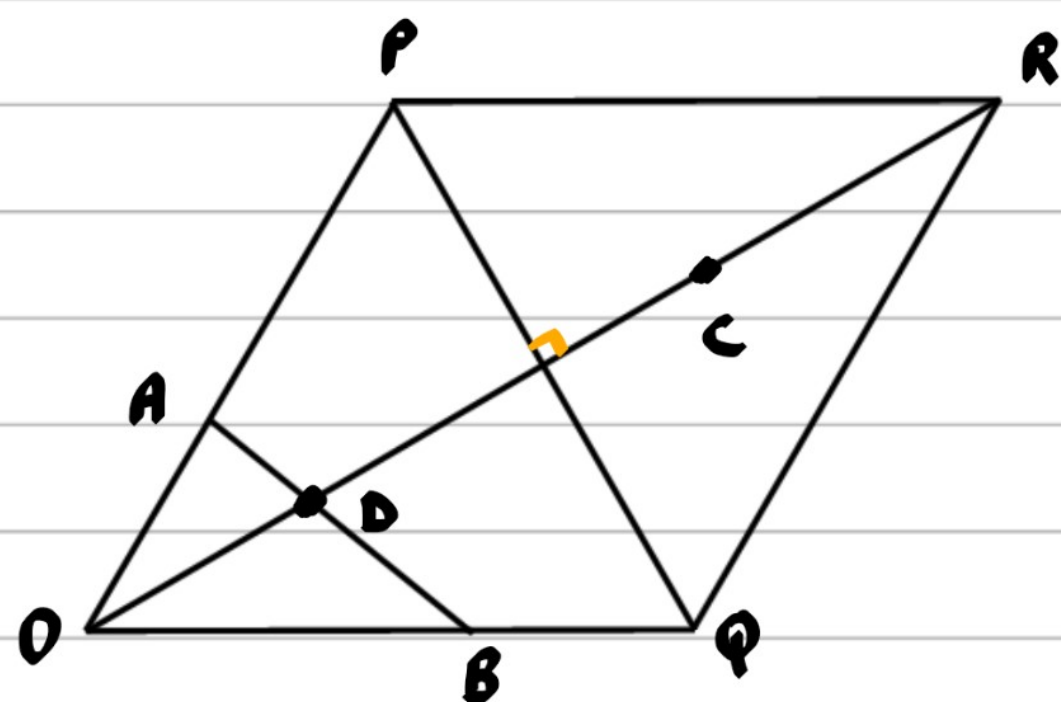
$\therefore \underline{b} - \underline{a}$ IS PERPENDICULAR TO $\underline{c}$

(b) $\vec{OD} = \vec{OA} + \vec{AD}$; $\vec{AD}$ IS A FRACTION OF $\vec{AB} \Rightarrow 0 < \lambda < 1$

$$\vec{OD} = \vec{OA} + \lambda \vec{AB}$$

$$\underline{d} = \underline{a} + \lambda (\underline{b} - \underline{a})$$

$\therefore \underline{d} = (1-\lambda)\underline{a} + \lambda\underline{b}$ AND $0 < \lambda < 1$ AS D IS IN BETWEEN A AND B



LET R BE THE POINT SUCH THAT $\vec{OP} + \vec{OQ} = \vec{OR}$

NOTE: $\vec{OQ}$ AND $\vec{PR}$ ARE PARALLEL AND EQUAL

$$\text{AS } |\vec{OP}| = |\vec{OQ}| \therefore |\vec{PR}| = |\vec{QR}|$$

$\therefore$ OPRQ IS A RHOMBUS

$\therefore$ DIAGONALS PQ AND OR ARE PERPENDICULAR

$\vec{OR}$ IS IN DIRECTION OF $\vec{OC}$

$$\therefore \underline{c} = m \vec{OR}$$

$$\therefore \underline{c} = m (\vec{OP} + \vec{OQ})$$



Question 5 continued

$$\therefore \underline{c} = m(\underline{b} + \underline{a})$$

AS ODC IS A STRAIGHT LINE $\therefore \vec{OD} = k \vec{OC}$

$$\therefore \vec{OD} = k * m(\underline{b} + \underline{a})$$

$$\therefore \vec{OD} = n(\underline{b} + \underline{a})$$

$$\therefore \underline{d} = n(\underline{b} + \underline{a})$$

FROM (b): $\underline{d} = (1 - \lambda)\underline{a} + \lambda\underline{b}$

$$\therefore (1 - \lambda)\underline{a} + \lambda\underline{b} = n(\underline{b} + \underline{a})$$

EQUATE COMPONENTS $a: 1 - \lambda = n$

$$b: \lambda = n \therefore n = \frac{\lambda}{a}$$

$$\therefore 1 - \lambda = \frac{\lambda}{a}$$

$$\therefore a - \lambda a = \lambda$$

$$\therefore a = \lambda a + \lambda$$

$$\therefore a = \lambda(a + 1)$$

$$\therefore \lambda = \frac{a}{a + 1}$$

$$OA = |\vec{OA}|$$

$$= a$$

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 5 continued

$$OB = |\vec{OB}|$$

$$= b$$

$$\therefore \frac{OA}{OB} = \frac{a}{b}$$

$$\vec{DA} = \vec{DO} + \vec{OA}$$

$$= -\underline{a} + \underline{a}$$

$$= -\{(1-\lambda)\underline{a} + \lambda\underline{b}\} + \underline{a}$$

$$= -(\underline{a} - \lambda\underline{a} + \lambda\underline{b}) + \underline{a}$$

$$= -\underline{a} + \lambda\underline{a} - \lambda\underline{b} + \underline{a}$$

$$= \lambda(\underline{a} - \underline{b})$$

$$DA = |\vec{DA}|$$

$$= \lambda |\underline{a} - \underline{b}|$$

$$= \frac{a}{a+b} |\underline{a} - \underline{b}|$$

$$\vec{DB} = \vec{DO} + \vec{OB}$$

$$= -\underline{a} + \underline{b}$$

$$= -\{(1-\lambda)\underline{a} + \lambda\underline{b}\} + \underline{b}$$

$$= -(\underline{a} - \lambda\underline{a} + \lambda\underline{b}) + \underline{b}$$

(Total for Question 5 is 16 marks)



$$= -\underline{a} + \lambda \underline{a} - \lambda \underline{b} + \underline{b}$$

$$= (\lambda - 1) \underline{a} + (1 - \lambda) \underline{b}$$

$$= (\underline{a} - \underline{b})(\lambda - 1)$$

$$DB = |\overrightarrow{DB}|$$

$$= |(\lambda - 1)| |\underline{a} - \underline{b}|$$

$$= \left| \left(\frac{a}{a+b} - 1 \right) \right| |\underline{a} - \underline{b}|$$

$$= \left| \frac{a - (a+b)}{a+b} \right| |\underline{a} - \underline{b}|$$

$$= \left| \frac{-b}{a+b} \right| |\underline{a} - \underline{b}|$$

$$= \frac{b}{a+b} |\underline{a} - \underline{b}|$$

$$\therefore \frac{DA}{DB} = \frac{\frac{a}{a+b} |\underline{a} - \underline{b}|}{\frac{b}{a+b} |\underline{a} - \underline{b}|} = \frac{a}{b}$$

$$\therefore \frac{DA}{DB} = \frac{OA}{OB} = \frac{a}{b}$$

$$\therefore DA : DB = OA : OB$$

6. Figure 1 shows a sketch of part of the curve with equation $y = x \sin(\ln x)$, $x \geq 1$

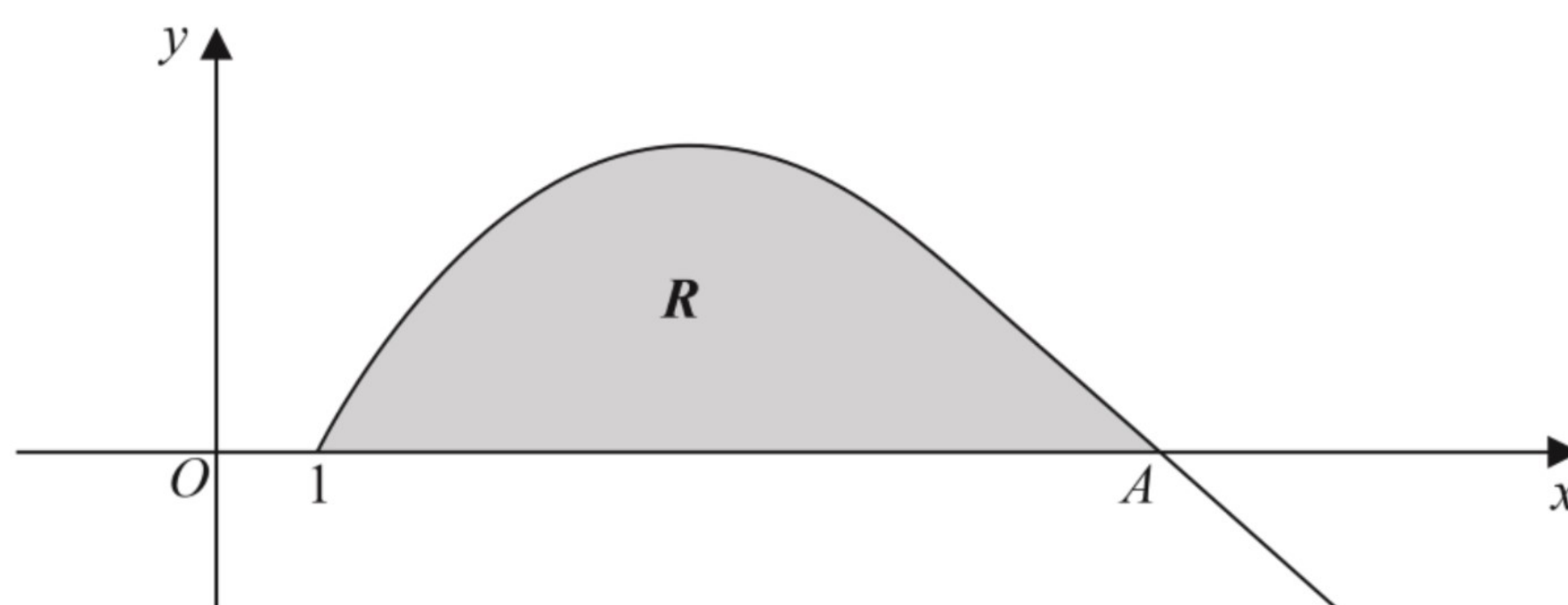


Figure 1

For $x > 1$, the curve first crosses the x -axis at the point A .

- (a) Find the x coordinate of A . (3)
- (b) Differentiate $x \sin(\ln x)$ and $x \cos(\ln x)$ with respect to x and hence find

$$\int \sin(\ln x) dx \text{ and } \int \cos(\ln x) dx \quad (7)$$

- (c) (i) Find $\int x \sin(\ln x) dx$.
- (ii) Hence show that the area of the shaded region R , bounded by the curve and the x -axis between the points $(1, 0)$ and A , is

$$\frac{1}{5}(e^{2\pi} + 1) \quad (9)$$

(a) $y = x \sin(\ln x)$

$$y = 0 : x \sin(\ln x) = 0$$

$$x \neq 0 \text{ As } x > 1$$

$$\sin(\ln x) = 0 \therefore \ln x = 0, \pi, 2\pi, \dots \therefore x = e^0, e^\pi, e^{2\pi}, \dots$$

$$\therefore x_A = e^\pi$$

(b) $y = x \sin(\ln x) \quad \therefore \frac{dy}{dx} = x \cdot \cos(\ln x) \cdot \frac{1}{x} + \sin(\ln x) \cdot 1$

Question 6 continued

$$= \cos(\ln x) + \sin(\ln x) \quad (2)$$

$$y = x \cos(\ln x) \quad (3) \quad \therefore \frac{dy}{dx} = x \cdot -\sin(\ln x) \cdot \frac{1}{x} + \cos(\ln x) \cdot 1$$

$$= \cos(\ln x) - \sin(\ln x) \quad (4)$$

$$(1) + (3): y = x \sin(\ln x) + x \cos(\ln x) \quad (5)$$

$$\therefore \frac{dy}{dx} = \{\cos(\ln x) + \sin(\ln x)\} + \{\cos(\ln x) - \sin(\ln x)\}$$

$$= 2 \cos(\ln x)$$

$$(1) - (3): y = x \sin(\ln x) - x \cos(\ln x) \quad (6)$$

$$\therefore \frac{dy}{dx} = \{\cos(\ln x) + \sin(\ln x)\} - \{\cos(\ln x) - \sin(\ln x)\}$$

$$= 2 \sin(\ln x)$$

$$\therefore \int \sin(\ln x) dx = \frac{1}{2} \{x \sin(\ln x) - x \cos(\ln x)\} + K_1$$

$$= \frac{1}{2} x \{\sin(\ln x) - \cos(\ln x)\} + K_1$$

$$\therefore \int \cos(\ln x) dx = \frac{1}{2} \{x \sin(\ln x) + x \cos(\ln x)\} + K_2$$

$$= \frac{1}{2} x \{\sin(\ln x) + \cos(\ln x)\} + K_2$$

$$(c)(i) \int x \sin(\ln x) dx$$

$$u = x$$

$$v = \frac{1}{2} x \{\sin(\ln x) - \cos(\ln x)\}$$

$$u' = 1$$

$$v' = \sin(\ln x)$$



Question 6 continued

$$\int x \sin(\ln x) dx = \frac{1}{2} x^2 \{ \sin(\ln x) - \cos(\ln x) \} -$$

$$\int \frac{1}{2} x \{ \sin(\ln x) - \cos(\ln x) \} dx$$

$$= \frac{1}{2} x^2 \{ \sin(\ln x) - \cos(\ln x) \} - \frac{1}{2} \int x \sin(\ln x)$$

$$+ \frac{1}{2} \int x \cos(\ln x) dx$$

$$\frac{3}{2} \int x \sin(\ln x) dx = \frac{1}{2} x^2 \{ \sin(\ln x) - \cos(\ln x) \} + \frac{1}{2} \int x \cos(\ln x) dx$$

$$3 \int x \sin(\ln x) dx = x^2 \{ \sin(\ln x) - \cos(\ln x) \} + \int x \cos(\ln x) dx \quad \textcircled{i}$$

NOTE: $\int x \cos(\ln x) dx$

$$u = x \quad v = \frac{1}{2} x \{ \sin(\ln x) + \cos(\ln x) \}$$

$$u' = 1 \quad v' = \cos(\ln x)$$

$$\int x \cos(\ln x) dx = \frac{1}{2} x^2 \{ \sin(\ln x) + \cos(\ln x) \}$$

$$- \int \frac{1}{2} x \{ \sin(\ln x) + \cos(\ln x) \} dx$$

$$= \frac{1}{2} x^2 \{ \sin(\ln x) + \cos(\ln x) \} - \frac{1}{2} \int x \sin(\ln x) dx$$

$$- \frac{1}{2} \int x \cos(\ln x) dx$$

$$\frac{3}{2} \int x \cos(\ln x) dx = \frac{1}{2} x^2 \{ \sin(\ln x) + \cos(\ln x) \} - \frac{1}{2} \int x \sin(\ln x) dx$$

$$3 \int x \cos(\ln x) dx = x^2 \{ \sin(\ln x) + \cos(\ln x) \} - \int x \sin(\ln x) dx \quad \textcircled{ii}$$

$$\textcircled{i} \times 3: 9 \int x \sin(\ln x) dx = 3x^2 \{ \sin(\ln x) - \cos(\ln x) \} + 3 \int x \cos(\ln x) dx \quad \textcircled{iii}$$

$$\textcircled{ii} \rightarrow \textcircled{iii}: 9 \int x \sin(\ln x) dx = 3x^2 \{ \sin(\ln x) - \cos(\ln x) \} + x^2 \{ \sin(\ln x) + \cos(\ln x) \}$$

$$- \int x \sin(\ln x) dx + K$$

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Question 6 continued

$$10 \int x \sin(\ln x) dx = 3x^2 \sin(\ln x) - 3x^2 \cos(\ln x) + x^2 \sin(\ln x)$$

$$+ x^2 \cos(\ln x) + K$$

$$= 4x^2 \sin(\ln x) - 2x^2 \cos(\ln x) + K$$

$$= 2x^2 \{2 \sin(\ln x) - \cos(\ln x)\} + K$$

$$\therefore \int x \sin(\ln x) dx = \frac{1}{5} x^2 \{2 \sin(\ln x) - \cos(\ln x)\} + K$$

(ii) AREA = $\int_1^{e^\pi} x \sin(\ln x) dx$

$$= \left[\frac{1}{5} x^2 \{2 \sin(\ln x) - \cos(\ln x)\} \right]_1^{e^\pi}$$

$$= \left[\frac{1}{5} (e^\pi)^2 \{2 \sin(\ln e^\pi) - \cos(\ln e^\pi)\} \right] - \left[\frac{1}{5} (1)^2 \{2 \sin(\ln 1) - \cos(\ln 1)\} \right]$$

$$= \frac{1}{5} e^{2\pi} (2 \sin \pi - \cos \pi) - \frac{1}{5} (2 \sin 0 - \cos 0)$$

$$= \frac{1}{5} e^{2\pi} (0 - (-1)) - \frac{1}{5} (0 - 1)$$

$$= \frac{1}{5} e^{2\pi} (1) - \frac{1}{5} (-1)$$

$$\therefore R = \frac{1}{5} (e^{2\pi} + 1) \text{ UNIT}^2$$

(Total for Question 6 is 19 marks)



7. Figure 2 shows a rectangular section of marshland, $OABC$, which is a metres long by b metres wide, where $a > b$.

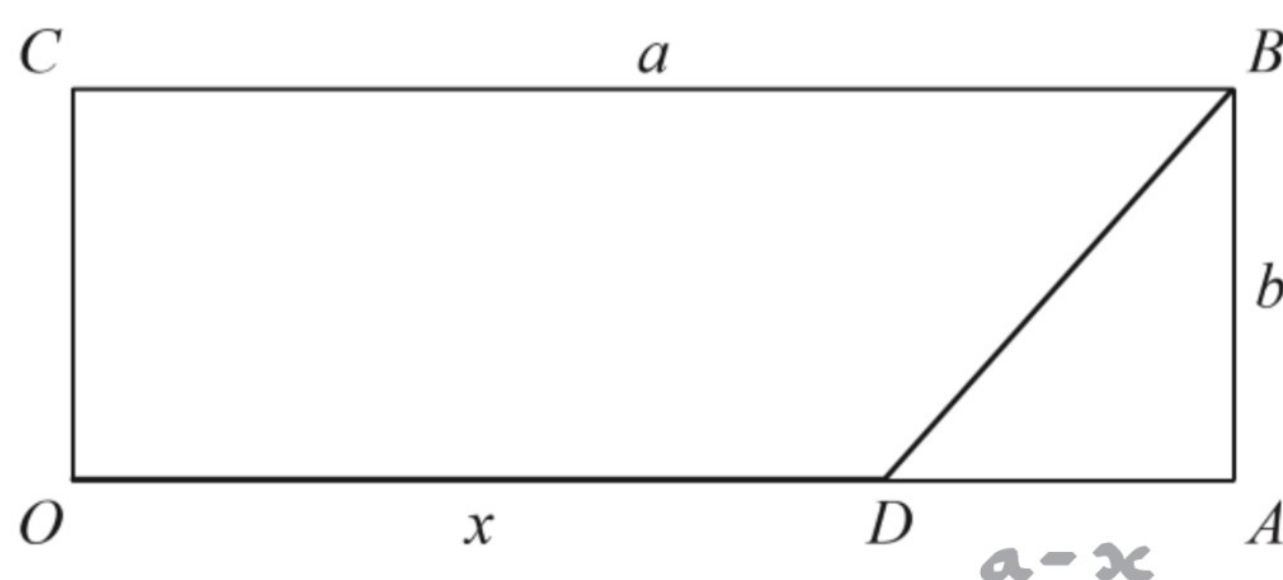


Figure 2

Edgar intends to get from O to B in the shortest possible time. In order to do this, he runs along edge OA for a distance x metres ($0 \leq x < a$) to the point D before wading through the marsh directly from D to B .

Edgar can wade through the marsh at a constant speed of 1 ms^{-1} , and he can run along the edge of the marsh at a constant speed of $\lambda \text{ ms}^{-1}$, where $\lambda > 1$

- (a) By finding an expression in terms of x for the time taken, t seconds, for Edgar to reach B from O , show that

$$\frac{dt}{dx} = \frac{1}{\lambda} - \frac{a-x}{\sqrt{(a-x)^2 + b^2}} \quad (5)$$

- (b) (i) Find, in terms of a , b and λ , the value of x for which $\frac{dt}{dx} = 0$

(ii) Show that this value of x lies in the interval $0 \leq x < a$ provided $\lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$

(iii) For λ in this range, show that the value of x found in (b)(i) gives a minimum value of t . (8)

- (c) Find the minimum time taken for Edgar to get from O to B if

(i) $\lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$

(ii) $1 < \lambda < \sqrt{1 + \frac{b^2}{a^2}}$ (4)

Edgar's friend, Frankie, also runs at a constant speed of $\lambda \text{ ms}^{-1}$. Frankie runs along the edges

OA and AB . Given that $\lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$

- (d) find the range of values of λ for which Frankie gets to B from O in a shorter time than Edgar's minimum time. (3)

(+S2)



Question 7 continued

(a)

$$DB = \sqrt{(a-x)^2 + b^2}$$

$$t = \frac{x}{\lambda} + \frac{\sqrt{(a-x)^2 + b^2}}{1}$$

$$t = \frac{x}{\lambda} + \{(a-x)^2 + b^2\}^{\frac{1}{2}}$$

$$\frac{dt}{dx} = \frac{1}{\lambda} + \frac{1}{2} \cdot \{(a-x)^2 + b^2\}^{-\frac{1}{2}} \cdot 2(a-x)' \cdot (-1)$$

$$\frac{dt}{dx} = \frac{1}{\lambda} - \frac{(a-x)}{\sqrt{(a-x)^2 + b^2}}$$

(b)(i)

$$\frac{dt}{dx} = 0 : \frac{1}{\lambda} - \frac{(a-x)}{\sqrt{(a-x)^2 + b^2}} = 0$$

$$\frac{(a-x)}{\sqrt{(a-x)^2 + b^2}} = \frac{1}{\lambda}$$

$$\lambda(a-x) = \sqrt{(a-x)^2 + b^2}$$

$$\lambda^2(a-x)^2 = (a-x)^2 + b^2$$

$$\lambda^2(a-x)^2 - (a-x)^2 = b^2$$

$$(a-x)^2(\lambda^2 - 1) = b^2$$

$$(a-x)^2 = \frac{b^2}{\lambda^2 - 1}$$

$$a-x = \pm \sqrt{\frac{b^2}{\lambda^2 - 1}}$$

$$a-x = \sqrt{\frac{b^2}{\lambda^2 - 1}} \quad (\text{As } x < a)$$



Question 7 continued

$$\therefore x = a - \frac{b}{\sqrt{\lambda^2 - 1}} \quad (\text{As } x < a)$$

(ii) For $x \geq 0$: $a - \frac{b}{\sqrt{\lambda^2 - 1}} \geq 0$

$$a \geq \frac{b}{\sqrt{\lambda^2 - 1}}$$

$$\sqrt{\lambda^2 - 1} \geq \frac{b}{a}$$

$$\lambda^2 - 1 \geq \frac{b^2}{a^2}$$

$$\lambda^2 \geq 1 + \frac{b^2}{a^2}$$

$$\therefore \lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$$

(iii) $\frac{dt}{dx} = \frac{1}{\lambda} = \frac{a-x}{\{(a-x)^2 + b^2\}^{\frac{1}{2}}}$

$$u = -(a-x) \quad v = \{(a-x)^2 + b^2\}^{\frac{1}{2}}$$

$$\begin{aligned} u' &= 1 & v' &= \frac{1}{2} \{(a-x)^2 + b^2\}^{-\frac{1}{2}} \cdot 2 \cdot (a-x)' \cdot (-1) \\ & & &= \frac{-(a-x)}{\{(a-x)^2 + b^2\}^{\frac{1}{2}}} \end{aligned}$$

$$\frac{d^2t}{dx^2} = \frac{\{(a-x)^2 + b^2\}^{\frac{1}{2}} \times 1 - \left[-(a-x) \times \frac{-(a-x)}{\{(a-x)^2 + b^2\}^{\frac{1}{2}}} \right]}{(\{(a-x)^2 + b^2\}^{\frac{1}{2}})^2}$$



Question 7 continued

$$= \frac{\{(a-x)^2 + b^2\}^{\frac{1}{2}} - \frac{(a-x)^2}{\{(a-x)^2 + b^2\}^{\frac{1}{2}}}}{\{(a-x)^2 + b^2\}}$$

$$= \frac{\{(a-x)^2 + b^2\}^{\frac{1}{2}} \{(a-x)^2 + b^2\}^{\frac{1}{2}} - (a-x)^2}{\{(a-x)^2 + b^2\} \{(a-x)^2 + b^2\}^{\frac{1}{2}}}$$

$$= \frac{(a-x)^2 + b^2 - (a-x)^2}{\{(a-x)^2 + b^2\}^{\frac{3}{2}}}$$

$$= \frac{b^2}{\{(a-x)^2 + b^2\}^{\frac{3}{2}}} > 0 \text{ FOR ALL } x \in \mathbb{R}$$

$\therefore$ minimum

(c)(i) $x = a - \frac{b}{\sqrt{\lambda^2 - 1}} : t = \frac{x}{\lambda} + \sqrt{(a-x)^2 + b^2}$

$$t = \frac{\left\{a - \frac{b}{\sqrt{\lambda^2 - 1}}\right\}}{\lambda} + \sqrt{\left\{a - \left(a - \frac{b}{\sqrt{\lambda^2 - 1}}\right)\right\}^2 + b^2}$$

$$t = \frac{a}{\lambda} - \frac{b}{\lambda \sqrt{\lambda^2 - 1}} + \sqrt{\left(\frac{b}{\sqrt{\lambda^2 - 1}}\right)^2 + b^2}$$

$$t = \frac{a}{\lambda} - \frac{b}{\lambda \sqrt{\lambda^2 - 1}} + \sqrt{\frac{b^2}{\lambda^2 - 1} + b^2}$$

$$t = \frac{a}{\lambda} - \frac{b}{\lambda \sqrt{\lambda^2 - 1}} + \sqrt{\frac{b^2 + b^2(\lambda^2 - 1)}{\lambda^2 - 1}}$$

$$t = \frac{a}{\lambda} - \frac{b}{\lambda \sqrt{\lambda^2 - 1}} + \sqrt{\frac{b^2 \lambda^2}{\lambda^2 - 1}}$$

$$t = \frac{a}{\lambda} - \frac{b}{\lambda \sqrt{\lambda^2 - 1}} + \frac{b\lambda}{\sqrt{\lambda^2 - 1}}$$



Question 7 continued

$$t = \frac{a}{\lambda} - \frac{b}{\lambda\sqrt{\lambda^2-1}} + \frac{\lambda^2 b}{\lambda\sqrt{\lambda^2-1}}$$

$$t = \frac{1}{\lambda} \left\{ a - \frac{b + \lambda^2 b}{\sqrt{\lambda^2-1}} \right\}$$

$$t = \frac{1}{\lambda} \left\{ a + \frac{b(\lambda^2-1)}{\sqrt{\lambda^2-1}} \right\}$$

$$\therefore t = \frac{a + b\sqrt{\lambda^2-1}}{\lambda}$$

(ii) To minimise t , As λ decreases, x decreases.

WE HAVE SHOWN THAT WHEN $\lambda = \sqrt{1 + \frac{b^2}{a^2}}$, $x = 0$

SINCE x IS A DISTANCE IT CANNOT BE NEGATIVE, SO ONCE IT GETS DOWN TO 0 IT STICKS AT THAT VALUE EVEN WHEN λ DECREASES MORE

SO FOR SMALLER VALUES OF λ , $x = 0$

$$t = \frac{0}{\lambda} + \sqrt{(a-0)^2 + b^2}$$

$$\therefore t = \sqrt{a^2 + b^2}$$

(d) $t_{\text{FRANKIE}} = \frac{a+b}{\lambda}$

$$t_{\text{EDGAR}} = \frac{a + b\sqrt{\lambda^2-1}}{\lambda}$$

(Total for Question 7 is 22 marks)

FOR STYLE, CLARITY AND PRESENTATION: 7 MARKS

TOTAL FOR PAPER: 100 MARKS

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$$t_{\text{FRANKIE}} < t_{\text{EDGAR}} : \frac{a+b}{\lambda} < \frac{a+b\sqrt{\lambda^2-1}}{\lambda}$$

$$a+b < a+b\sqrt{\lambda^2-1}$$

$$b < b\sqrt{\lambda^2-1}$$

$$1 < \sqrt{\lambda^2-1}$$

$$1 < \lambda^2 - 1$$

$$2 < \lambda^2$$

$$\therefore \lambda > \sqrt{2} \text{ ONLY}$$

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